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Effects of Demographic Characteristics on Safety Climate and Construction Worker Safety Behavior

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Abstract: Safety climate and safety behavior have been proven to be critical for construction safety. However, few studies have systematically tested the influences of demographic characteristics on construction worker safety, especially in the background of China's construction industry. There is still a gap in the demographics–safety relationship, which may reduce the relevance and effectiveness of safety measures. This study explored the effects of five demographic factors (gender, age, work experience, education level, and employer size) on two safety indicators (safety climate and construction worker safety behavior). The survey data were collected from Chinese construction workers. The *t*-test and one-way ANOVA methods were adopted to examine the demographic differences. The results show that all five demographic factors have significant effects on safety climate, while three (age, education level, and employer size) have significant influences on construction worker safety behavior. However, the remaining two demographic factors (gender and work experience) do not show significant effects on construction worker safety behavior. The results remind the safety management personnel to take a pragmatic approach and to formulate safety measures based on the demographic characteristics of different worker sub-groups.

Keywords: demographic characteristic; safety climate; construction worker safety behavior; sub-group analysis

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1. Introduction

Despite its best efforts, the construction industry continues to be one of the most dangerous industries worldwide [1–3]. Due to rapid urbanization and large-scale infrastructure construction, China's construction industry also suffers from many fatalities and accidents. From 2009 to 2017, 5205 fatal accidents occurred in China's housing construction and municipal engineering projects, resulting in 6354 fatalities [4]. Most of the 55.58 million construction workers in China's 2021 construction industry are peasant workers [5]. They usually have poor living conditions, are separated from their families for a long time, have a low sense of belonging to the local cities, and rarely participate in community activities or trade unions. Of the peasant workers, 38.8% are over 50 years old, and 52.4% possess middle school or below education [6]. These numbers show that there is still much room for China's construction industry to further improve its overall safety performance, and it is urgent to strengthen the safety competence of construction workers because of their general demographic and social characteristics.

Some demographic factors (such as education level, age, and ethnicity) could affect safety climate [7] and safety behavior [8], which are critical to worker safety. There are many examples of projects with poor safety performance due to managers ignoring the influence of demographic characteristics of workers. However, systematic studies about

this issue are still rare. To address this problem, this study tries to explore the demographic effect on worker safety from a holistic perspective. Several researchers studied the factors related to construction worker safety and showed that frontline workers play a critical role in improving safety performance and safety outcomes [4,7,9–11]. Among the safety factors, safety climate was considered a vital antecedent of construction worker safety [12–21]. As a preventive indicator, safety behavior is typically chosen to measure safety performance [22–24]. There exist studies to explore the relationships between construction worker safety behavior and its antecedents with the help of various methods, such as correlation analysis [7], regression analysis [10], structural equation model [25], belief neural network [26], and agent-based simulation [8]. However, few studies have systematically tested the influences of demographic characteristics on construction worker safety, especially in the background of China's construction industry. Some studies explored the impacts of ethnicity on safety climate [7], the influences of demographic factors on construction employees' safety perceptions [9] and safety consciousness, and the safety citizenship behavior of construction workers [11]. However, these studies rarely measured the systematical effects of demographic characteristics on construction worker safety, such as safety climate and safety behavior. That is to say, although there are studies related to the effects of demographic characteristics on safety outcomes [10] or safety climate [27] and the effects of safety climate on safety behaviors and safety outcomes [28–30], it is still unknown what the effects of demographic characteristics on safety behavior are, and it is also unknown what the differentiated results of the safety climate and safety behavior are for different groups of workers under the influence of demographic factors, especially under the circumstance of Chinese construction industry. Thus, a multivariate study related to assessing multiple demographic effects on safety climate and safety behavior needs deeper exploration.

This research gap may reduce the effectiveness of safety training and reduce the ability to accurately intervene in specific cohorts of construction workers. For example, many safety trainings cover all construction workers. Managers do not know the psychological characteristics, intensity of safety needs, and safety skill differences of workers with different ages and education levels, thus reducing the training effectiveness. Meanwhile, contractors of different sizes have different levels of safety investment, which may affect their safety climate and safety performance. Thus, this research helps to fill this gap and make it clear how demographic variables (gender, age, education level, work experience, and employer size) can affect construction worker safety (safety climate, safety behavior). This study aims to clarify: (1) whether demographic characteristics (gender, age, education level, work experience, and employer size) affect the levels of safety climate and safety behavior of construction workers and (2) which specific construction worker cohorts behave worse and need more attention. The *t*-test and one-way ANOVA methods were adopted to examine the demographic effects. This study may help the safety management personnel to take a pragmatic approach and formulate safety measures based on the demographic characteristics of different worker sub-groups. Further, this research can contribute to fostering more studies related to demographic and organizational effects on construction personnel safety.

2. Literature Review

2.1. Safety Climate and Safety Behavior

There are many theories that help to set up the relationships between demographics, safety climate, and safety behavior, such as the theoretical framework for the psychosocial decision-making process of construction worker safety behavior [8], Human Factors Analysis and Classification System (HFACS) [26], and leadership-culture-behavior model [28]. Safety climate refers to the commonly accepted perceptions of beliefs, values, and procedures about organizational safety [31], with which the level of construction safety performance can be improved [32–34], and the number of accidents and injuries can be reduced [22,35]. As a leading indicator of safety performance [36], safety behavior is de-

fined as the employee actions related to observing the organization's safety procedures [37], which can be further divided into two basic constructs: safety compliance and safety participation [38]. The former is defined as the in-role safety actions related to observing safety regulations and instructions, while the latter refers to the out-role safety behaviors related to reminding coworkers, voicing behavior, and so on [38,39].

Many studies confirmed a strong correlation between safety climate and safety behavior [21,40]. Safety climate can affect safety behavior in three ways: directly [18,25], indirectly through mediating variables such as stress and psychological capital [4,41], and indirectly through moderating variables such as work-group identity and site layout [12,14]. Thus, safety climate and safety behavior are twin constructs. Safety climate determines personal psychological states such as intention and attitude [8,21,40] and triggers safety behaviors of individual employees, which may impact organizational safety climate and group-level safety behaviors. In light of previous research findings, the safety climate and safety behavior are chosen to represent construction worker safety in this study as they can reflect the safety performance of workers at both individual and organizational levels.

2.2. Gender

There is no consensus about gender differences in worker safety. Some studies found that females behave better than males. Females have fewer accidents at work [42], report a lower frequency of work injuries [43], and can deal with more workplace hazards than males [44]. The possible reasons include that women are more cautious about potential risks [38], have greater concern for safety issues [45], and behave in a safe and attentive way when working than men.

Other studies found that females perform worse than males, such as females having more work-related injuries [46] and a lower level of safety citizenship behavior than males [11]. The possible causes may include the following factors: males may have a higher level of personal loyalty, masculinity [47], and interpersonal relationships [48], performing a more positive safety citizenship behavior. Safety equipment or work process are typically better suited to males than females [49], and females confront the problems of gender discrimination and sexual harassment [50], which may reduce the safety performance of females.

Finally, some concluded that there are no significant gender differences in personal safety at work, such as safety proactivity behavior, hazard identification [51,52], and the frequency of microaccidents [53].

Factors such as safety management [52], safety neglect [53], and safety perceptions [9] may act simultaneously on personal safety and lead to different results. Thus, this study attempts to clarify which gender differences relate to safety climate and safety behavior.

2.3. Education

Education can affect worker safety. Workers with higher education backgrounds show better safety citizenship behavior and safety initiative [11,52]. There are some potential reasons leading to this result. Workers with high education levels hold stronger safety consciousness [11] and safety attitudes [54]. Further, highly educated workers may exhibit better perceptions of safety hazards [52] and interpersonal relationships [9] and possess better safety knowledge and skills to involve in their activities [7] and avoid possible risks [55,56]. In other words, education level can impact the internal safety motivation and ability of frontline workers and, in turn, promote their external safety performance.

However, the effect of education is not conclusive. In some studies, no significant relationships were found between education level and safety attitude, safety performance [57], or accident risk [58]. Another found that the power of education on safety behavior is weak and cannot bring obvious improvement in worker safety [21]. This study attempts to clarify the degree to which education affects safety climate and safety behavior.

2.4. Age and Work Experience

Age and experience are closely related, so they are discussed together here. For age, it is generally accepted that age is positively related to safety attitude, safety perception, and safety behavior [54,59] and negatively associated with injuries and events [60]. Older workers typically exhibit better safety than young workers [61]. However, uncertain relationships between age and worker safety were also reported. A review study showed that young construction workers may have a lower, same, or higher fatality rate than elder workers [62]. A U-shape age-safety relationship was identified that may explain this phenomenon [8], and middle-aged construction workers were prone to underestimate safety hazards and behave worse than workers of other age groups. The age-related physical conditions, work experiences, and safety attitudes may act on safety at the same time and cause this difference [63–65].

Research on the relationship between work experience and safety performance is also not conclusive. Some studies indicated that work experience helps to reduce injury rates and non-fatal accidents [66,67], while another found that midcareer workers showed worse safety performance than those with less experience [10]. Another two studies stated that neither work experience nor job tenure was associated with significant differences in accident risk, safety proactivity behaviors, and safety hazard identification [52,58]. The sensitivity of work experience to enhance safety behavior was also proved to be low [21]. Thus, it is necessary to further investigate and sort out the influence of age and work experience on worker safety.

2.5. Employer Size

Researchers found that employer size can affect both individual and organizational safety performance. Employees in microfirms have a lower use rate of personal protective equipment and general protection measures than those in large firms [68]. Small and medium enterprises and microfirms exhibit a higher injury rate and experience more dangerous work sites than large firms [69]. There are some possible causes contributing to these differences. First, large firms possess abundant resources [70] and well-developed safety management systems [71], while small and medium enterprises have limited resources and find it difficult to set up and utilize safety management systems. Second, top managers of small and medium enterprises focus more on tasks than safety [70] and are reluctant to invest in safety when it may appear to diminish profits or growth potential [72], which may impact the levels of safety training and safety knowledge for their employees [52]. Third, workers of general contractors hold a better safety perception [59] and a higher overall mean safety climate score than subcontractor workers [7]. For China's construction industry, many subcontractors are small and medium enterprises. The safety performance of construction workers in these enterprises may be influenced by their employers' size.

3. Methodology

3.1. Research Design

The data collection and analysis method consists of four steps: first, through literature analysis, the researchers identify the important demographic variables that may affect the safety climate and safety behavior; second, the classification methods of demographic variables were sorted out, and the measurement indicators of safety climate and safety behavior were selected, forming a preliminary survey questionnaire; third, safety experts from universities and senior safety managers were invited to evaluate the questionnaire and modify it in terms of readability, popularity, and accuracy to ensure its validity; finally, based on the modified questionnaire, site surveys were conducted to obtain the survey data, and the SPSS software program was used for data analysis.

3.2. Participants

The participants came from 22 Chinese construction projects. With the help of site managers, 536 valid questionnaires filled out by construction workers were collected. The

surveys were carried out during their safety training, toolbox meetings, or work shift changes. The participants were told to finish the questionnaires voluntarily and independently based on their true feelings. Their private information would not be collected. The survey data was kept confidential strictly and only used in academic research. Of the 536 respondents, 91.60% (491) are male, and 8.4% (45) are female. As to age, 75 respondents are less than 30 years old, 220 respondents are 30 to 39, 176 respondents are 40–49, and 65 respondents are 50 or older. For the education level, 117 participants have an elementary school degree, 284 participants have a middle school degree, and 135 participants have a high school degree or higher. In terms of work experience background, 132 respondents work less than 5 years, 200 respondents 5–9 years, 126 respondents 10–14 years, and 78 respondents 15 years or more. As to the employer size, 76 participants work in an organization with fewer than 20 persons, 340 participants in an organization with 21–300 persons, and 120 participants in an organization with more than 300 persons.

3.3. Measures

3.3.1. Demographic Information

The first part of the questionnaire was demographic information. The participants were asked about their gender, age, education level, work experience, and employer size. These information sets were saved in a standard format and used in the data analysis.

3.3.2. Safety Climate

Safety climate was measured by asking participants to indicate whether they agreed with statements using a 5-point Likert scale where 1 stands for "strongly disagree" and 5 stands for "strongly agree." The average value of the responses was utilized to score each dimension, which is suitable for measuring a global construct and obtaining a better concept generality [73–75]. The statements were associated with sub-dimensions of management commitment (6 statements, e.g., "Our management provides safe equipment"), supervisor perception (6 statements, e.g., "My supervisor keeps workers informed of safety rules"), coworker perception (4 statements, e.g., "My coworkers encourage others to be safe"), work pressure (2 items, e.g., "There are enough workers to carry out the required work"), and role overload (2 statements, e.g., "I am so busy on the job that I can't take normal breaks"). This five-dimension measurement of safety climate has been employed in related studies, and its validity and reliability of the measurement scales have been illustrated [4,10,32].

3.3.3. Safety Behavior

Safety behavior was assessed with sub-dimensions of safety compliance (3 statements, e.g., " I use all the necessary safety equipment to do my job"), and safety participation (3 statements, e.g., " I put in extra effort to improve the safety of the workplace") which were developed by Neal and Griffin [76]. The same 5-point Likert scale and averaged score as in safety climate was adopted to evaluate the sub-dimensions. The questionnaire is shown in Appendix A.

3.4. Data Analysis

This study examines the impacts of demographic characteristics (gender, age, education level, work experience, and employer size) on construction worker safety (safety climate and safety behavior). Among the five characteristics, gender has two categories, and the remaining four characteristics have three or more sub-groups. Therefore, an independent-sample *t*-test is used to analyze the impacts of gender, and a one-way analysis of variance (ANOVA) is adopted to explore the influences of other four variables (age, education level, work experience, and employer size).

The analysis process is as follows. First, test the homogeneity of variance of the data. If p is higher than 0.05, it can be considered that the variance is homogeneous; otherwise, it is concluded that the variance is not homogeneous. Second, we analyzed the data using the *t*-test or one-way ANOVA. For the *t*-test, if p is higher than 0.05, there is no significant

difference between the two data groups; and if p is lower than 0.05, it is inferred that there is a significant difference between the two groups. For one-way ANOVA, if the variance is homogeneous, the Scheffe result is used to compare the mean values of the data groups; if the variance is not homogeneous, the Tamhane result is used to judge the mean values. The above methods are used to analyze the impacts of demographic characteristics on construction worker safety. Due to the large number of demographic variables analyzed, only the analysis results with significant differences are shown herein.

3.5. Reliability and Validity Analyses

Reliability and validity analyses were carried out to ensure the effectiveness of the questionnaire. As to reliability, the Cronbach value was adopted, and its threshold value was 0.7. The test results show that Cronbach values of safety climate and safety behavior were all above 0.7, which indicates that the questionnaire had good convergent reliability.

For validity, standard factor loadings (SFL), construct reliability (CR), and average variance extracted (AVE) were used, and their threshold values were 0.5, 0.7, and 0.5, respectively. The results prove that all SFLs and CRs of safety climate and safety behavior were higher than 0.5 and 0.7, respectively. AVEs of safety climate and safety behavior were close to 0.5 and higher than 0.7, respectively, which illustrates a good convergent validity of the questionnaire.

4. Results

4.1. Overall Effects of Demographic Characteristics on Safety Climate and Safety Behavior

The overall effects of demographic characteristics on safety climate and safety behavior were tested with the method of ANOVA. The results are shown in Table 1. It can be seen that employer size has significant effects on the overall constructs of both safety climate and safety behavior, while age and education level have significant effects on safety behavior. The results indicate that most of the differences among demographic indicators and the overall constructs of safety climate and safety behavior are not statistically significant, illustrating the necessity of subgroup analysis.

Table 1. Difference analysis of SC and SB by demographic variables.

	Gender									
	Male				Female				F	p
	Mean		Std		Mean		Std			
SC	3.74		0.52		3.60		0.51		3.321	0.069
SB	3.92		0.77		3.84		0.57		0.449	0.503
	Age									
	<30 years		30–39 years		40–49 years		≥50 years		F	p
	Mean	Std	Mean	Std	Mean	Std	Mean	Std		
SC	3.78	0.54	3.76	0.54	3.69	0.46	3.66	0.48	1.295	0.275
SB	4.12	0.65	3.91	0.77	3.75	0.79	4.09	0.58	5.851	**0.001**
	Education level									
	Elementary school		Middle school		High school or higher				F	p
	Mean	Std	Mean	Std	Mean		Std			
SC	3.66	0.41	3.72	0.53	3.81		0.54		2.648	0.072
SB	3.87	0.66	3.83	0.80	4.10		0.68		6.366	**0.002**
	Work experience									
	<5 years		5–9 years		10–14 years		≥15 years		F	p
	Mean	Std	Mean	Std	Mean	Std	Mean	Std		
SC	3.71	0.45	3.70	0.52	3.70	0.54	3.87	0.53	2.385	0.068
SB	3.93	0.67	3.89	0.78	3.84	0.74	4.03	0.83	1.122	0.340

Table 1. *Cont.*

	Gender							
	Employer size							
	1–20 employees		21–300 employees		300 or more employees		F	*p*
	Mean	Std	Mean	Std	Mean	Std		
SC	3.62	0.77	3.69	0.46	3.90	0.42	9.350	**0.000**
SB	3.68	0.99	3.94	0.67	3.96	0.78	4.102	**0.017**

Note: SC = safety climate; SB = safety behavior; bold values indicate significant differences.

4.2. Effects of Demographic Characteristics on Subdimensions of Safety Climate

(1) Gender

Gender has no significant impact on management commitment, supervisor perception, work pressure, and role overload but has a significant impact on coworker perception. Male workers' perception of coworkers is significantly higher than female workers' (Tables A1 and A2).

(2) Age

Only age has a significant impact on management commitment. Workers aged under 30 have the highest level of perceived management commitment, while workers aged 50 or over have the lowest level of perceived management commitment, and the former is significantly higher than the latter. The perception level of management commitment for workers aged 30–49 is medium, and there is no significant difference with that for workers aged 50 or over (Tables A3 and A4).

(3) Education level

Education level has a significant impact on coworker perception but no other latent variables. The coworker perception for workers with an education level of high school or higher is significantly greater than that for workers with just elementary school. Further, the higher the education level is, the higher the coworker perception level is (Tables A5 and A6).

(4) Work experience

Work experience has a significant impact on management commitment but not on the other variables. However, workers with different years of experience have little difference in their perceptions of management commitment (Tables A7 and A8).

(5) Employer size

Employer size has a significant impact on the safety climate. The larger the employer size is, the higher the levels of management commitment, supervisor perception, and coworker perception. The levels of work pressure and role overload for workers in organizations with 21–300 people are the highest and are significantly higher than those for workers in organizations with more than 300 people. The levels of work pressure and role overload for workers in organizations with 1–20 people are lower than those for workers in medium-sized organizations but higher than those for workers in large organizations (Tables A9 and A10).

The results show that gender and education level are significantly associated with coworker conception; age and work experience are significantly related to management commitment; and employer size is significantly correlated with both five sub-dimensions of safety climate (management commitment, supervisor perception, coworker perception, work pressure, and role overload).

4.3. Effects of Demographic Characteristics on Subdimensions of Safety Behavior

(1) Age

Age has a significant impact on safety behavior. Construction workers under the age of 30 have the highest level of safety compliance, while construction workers aged 40–49 have the lowest level of safety compliance and behave significantly worse than construction workers under the age of 40. Construction workers aged 50 or over have the highest level of safety participation, while construction workers aged 40–49 have the lowest level of

safety participation and behave significantly worse than workers under 30 and over 50 (Tables A11 and A12).

(2) Education level

Education level has no significant impact on safety compliance but has a significant impact on safety participation. Construction workers with education levels of high school or higher have the highest level of safety participation, followed by construction workers with elementary school education, and construction workers with middle school education have the lowest level of safety participation (Tables A13 and A14).

(3) Employer size

Employer size has a significant impact on safety compliance and safety participation, but the differences between sub-groups are small. Workers in large organizations have the highest level of safety compliance, workers in medium organizations have the highest level of safety participation, and workers in small organizations have the lowest level of both safety compliance and safety participation (Tables A15 and A16).

Overall, age, education level, and employer size have significant impacts on the safety behavior of construction workers. The safety compliance level of construction workers under 30 in large organizations is the highest, while the safety participation level of construction workers over 50 in medium organizations is the highest. Further, the safety compliance level and safety participation level of construction workers aged 40–49 in small organizations are the lowest. The summarized results are shown in Table 2 as follows. In Table 2, the affecting factors (five demographic characteristics) were listed in the first column, and the affected factors (seven safety climate and safety behavior indicators) were listed in other columns. The tick means that the impact is significant.

Table 2. The results about the effects of demographic characteristics on safety climate and safety behavior.

Demographic Characteristics	Safety Climate					Safety Behavior	
	Management Commitment	Supervisor Perception	Coworker Perception	Work Pressure	Role Overload	Safety Compliance	Safety Participation
Gender			√				
Age	√					√	√
Education level			√				√
Work experience	√						
Employer size	√	√	√	√	√	√	√

Note: √ = significant impact.

5. Discussion

5.1. Major Findings

This study explored the impacts of five demographic factors on safety climate and construction worker safety behavior in the Chinese construction industry. The *t*-test and ANOVA methods were adopted to detect the differences in these two safety indicators derived from gender, age, education level, work experience, and employer size. The results indicate that certain components of safety climate are significantly correlated with five demographic factors (age, gender, education level, work experience, and employer size). Specifically, coworker perception is significantly correlated with gender and education level; management commitment is significantly correlated with age and work experience; five sub-dimensions of safety climate are significantly correlated with employer size. Meanwhile, certain components of construction worker safety behavior are significantly affected by age, education level, and employer size. In contrast, construction worker safety behavior is not significantly impacted by gender and work experience. Since more than 90% of the respondents are male, and more than 60% of the respondents have less than 10 years of work experience, the impacts of gender and work experience on safety behavior may not be apparent.

(1) Effects of age on safety climate and safety behavior

Age has a significant negative impact on management commitment. In other words, younger workers have a higher perception level of management commitment than older workers. It is possible that young workers are more curious about site conditions and know less about safety requirements, so they have a higher evaluation of safety management. By contrast, the older workers have worked at many sites and may compare the safety performance of different sites, which can reduce their evaluation level of safety management in their current sites. On the other side, construction workers aged 40–49 have the lowest levels of both safety compliance behavior and safety participation behavior, while workers aged under 30 and 50 or over have the highest levels of safety compliance behavior and safety participation behavior, respectively. Thus, a U-shape age-safety relationship may exist, which echoes Peng and Chan [64] and He et al. [8]. The finding that workers aged 50 or over have the highest levels of safety participation behavior is also consistent with the results of Wang et al. [52]. The workers aged 40–49 may become complacent [10] and actively lower their safety attitudes and safety awareness [63,65], which can reduce their safety behavior performance.

(2) Effects of work experience on safety climate and safety behavior

Work experience has a significant influence on management commitment, but the sub-group differences in management commitment are weak. Further, the result that work experience does not have significant impacts on safety behavior is consistent with [21,52,58]. There may be two reasons for this result: first, work experience may not have a significant relationship with safety behavior, but it may affect other safety-related variables (such as sustained attention) [58], and when combined with other variables (such as safety training, employee's involvement, safety management systems, and procedures) [21], it can significantly affect safety behavior, meaning that work experience is similar to a catalyst for safety behavior and cannot play a role alone, but has to rely on other factors to jointly influence safety behavior; second, with the continuous improvement of safety regulations, the safety training for construction workers at various construction sites is relatively standardized, and the safety competence among workers after training is not significantly different, thereby not reflecting the significant impact of work experience on safety behavior. The results on the impacts of age and work experience on worker safety remind safety managers to encourage middle-aged or middle-experienced workers to observe safety operation regulations, participate in safety activities, and voice their safety advice.

(3) Effects of gender on safety climate and safety behavior

Gender has a significant impact on coworker perception. Male workers' coworker perception is significantly better than female workers'. The possible reasons may be that males possess a higher level of safety voice [53], personal loyalty [47], and interpersonal relationships [48] and, thus, have a better attitude toward their coworkers. However, gender has no significant effects on other sub-dimensions of safety climate. The reason may be that in this survey, males accounted for the majority of the respondents (91.60%), so the sample size of female survey results is relatively small, and the survey data is easily influenced by certain values, resulting in insignificant results. Another possible reason is that male and female workers experience similar safety climates on site, so there is no significant difference for other sub-dimensions of safety climate. Further, gender also has no significant influence on safety behavior, which is consistent with [51,52]. The results confirm that gender differences in different safety indicators is inconclusive. With the gradual acceptance of the concept of gender equality, many gender-related safety studies are still to be explored in the male-dominated construction engineering community.

(4) Effects of education on safety climate and safety behavior

Education level has a significant positive influence on coworker perception, which is consistent with Han et al. [9] that highly educated workers possess better interpersonal relationships. Workers with higher education levels have better communication competence and higher self-confidence and are more willing to interact with their coworkers, thereby leading to a better coworker perception than those with lower education levels.

Meanwhile, education level has a significant impact on safety participation behavior, which is in accordance with the results of some related studies [11,52]. It is easy to accept that highly educated workers possibly hold higher levels of safety consciousness [11], safety attitude [54], and knowledge about safety activities [7], so they are more self-motivated and can maintain a higher level of safety participation.

(5) Effects of employer size on safety climate and safety behavior

Employer size has a significant positive effect on three sub-dimensions of safety climate (management commitment, supervisor perception, and coworker perception), which is consistent with the results of Chan et al. [7]. However, there may be an inverted U-shape relationship between employer size and the other two sub-dimensions of safety climate (work pressure and role overload). That is to say; medium-sized organizations have the highest levels of work pressure and role overload. This result may be derived from the mismatch between tasks and resources [70]. Most medium-sized organizations are subcontractors, and they undertake most of the production tasks and the first-line safety control responsibility. However, they may not have enough resources to simultaneously meet the demands of the general contractors (large organizations) and work teams (small organizations). This tension may cause high perceptions of psychological stress and role overload for workers. Meanwhile, employer size significantly affects both safety compliance behavior and safety participation behavior, and workers in small organizations have the lowest level of safety behavior, which echoes the findings of Boustras and Hadjimanolis [68]. For small construction organizations, poor safety management systems [71], insufficient safety investment [68], and deficient safety training [52] may all contribute to this result.

5.2. Theoretical Implications

This study possesses several theoretical contributions to the safety research domain. First, this study adds empirical evidence about the demographic differences in construction worker safety and systematically tests the demographic influences on safety climate and worker safety behavior in construction. Although there have been studies on the impact of demographic factors on safety, they generally only focus on analyzing one or two factors. There are few studies systematically analyzing five demographic factors such as this study. In addition, previous studies only took a certain safety indicator (such as safety citizenship behavior or safety perception) as the object of influence and rarely took both safety climate and safety behavior as the accompanying object of influence. This study analyzes these two safety indicators under the influences of five demographic factors, which may help to find the key demographic factors and the real affected safety objects.

Second, this study shows that there may exist several nonlinear relationships between demographic factors and construction worker safety. These relationships demonstrate the necessity of appropriate and targeted management measures. The nonlinear relationships may help to change the stereotype that a certain demographic factor can only have a linear impact on safety performance, thus providing a theoretical basis for developing safety training plans and safety management measures more scientifically. For example, previous studies generally believed that younger new employees were the group with a high incidence of safety problems. However, this study shows that construction workers aged 40–49 may face more social pressure, which can cause more safety problems, so they need to be paid additional attentions.

Third, the method that explores the effects of five demographic factors on the sub-dimensions of two safety indicators for the sub-groups of construction workers helps to clarify the real causes underpinning the safety performance and pinpoint the specific workgroups that need to be further improved. In previous relevant studies, whether a certain demographic factor has an impact on a certain safety indicator was usually analyzed, but which dimensions of the safety indicator are significantly affected were not explained in depth, nor were the differences of the significantly affected dimensions under a certain demographic factor between different cohorts pointed out, thereby reducing the theoretical value of the research results. This study not only tests whether a certain

demographic factor has an impact on a certain safety indicator but also indicates the impact on different dimensions of this indicator and verifies the differences between different cohorts of construction workers, thereby improving the theoretical depth and the research value.

5.3. Practical Implications

This study also has some practical implications. First, construction workers aged 40–49 should be given more attention, as their safety behaviors tend to be the worst compared to the workers of other age groups. Active measures should be taken to eliminate their job burnout, enhance their enthusiasm to participate in safety activities and improve their safety compliance and safety participation. Although the general research believes that construction workers aged 40–49 have more vigorous energy and richer work experience, they also have to bear various social and economic pressures, such as supporting their parents and paying for their children's education costs. At the same time, they also face psychological pressures such as job instability, alienation from their families, and job burnout. Therefore, it is necessary to take comprehensive measures for these workers to alleviate their various pressures.

Second, safety managers should be careful with workers with 5–9 years of work experience since their perceptions of management commitment are lower than other workers. More communication frequency, better leader–member exchange, and stronger trust relationships may help to change this situation. Workers with 5–9 years of work experience have generally experienced multiple construction projects and will compare the safety conditions and management styles under different projects. Once they find that the current project status is worse than the previous project or lower than expected, it is easy to have a psychological gap and reduce their perceptions of management commitment. At the same time, these workers may also have long-term plans for career development. Once they find that this plan is difficult to realize, they are also prone to have psychological problems such as disappointment and anxiety, which will affect their perceptions of the safety climate. Therefore, providing targeted psychological counseling and achievable career development guidance for this group can help improve the overall safety performance of the projects.

Third, safety personnel working in medium-sized organizations should reasonably arrange work tasks and human resources to reduce work pressure, while those working in small- or micro-sized organizations should establish sound safety management systems, put in sufficient safety investments, and provide adequate safety training to effectively improve the safety behavior performance of construction workers. This study found that workers with medium-sized employers have the lowest level of management commitment, supervisor perception, and coworker perception and face the highest level of role overload and work pressure than workers with large- or small-sized employers. Further, the workers of small contractors have the lowest safety behavior performance than those of medium or large contractors. On the other hand, most of China's construction workers are serving small- and medium-sized construction contractors whose technical ability, financial strength, and management level are relatively weak. Therefore, it is an effective measure to increase the support for small- and medium-sized construction contractors, reduce their business risks, and improve their management ability and strength for construction workers to improve the safety performance of construction workers in these small- and medium-sized enterprises.

5.4. Limitations and Future Study Directions

This study has some limitations due to the constraints of research resources and time. First, some other demographic characteristics, such as ethnic group or trade, could be further analyzed to explore their impacts on worker safety. Second, some other important safety indicators, such as unsafe events or accidents, may also be included in the future to indicate their outcomes under the effects of demographic characteristics. Third, more survey data are needed for future comparisons of how results differ by location, and

longitude studies are welcomed to test the stability of these results over time. Fourth, more research can further explore the indirect effects of demographic factors on safety behavior, such as the mediating effect of safety climate and the moderating effect of safety leadership in the demographics–safety relationship. Thus, a whole model connecting demographic characteristics, safety climate, and safety behavior may help to better reveal the inter-relationship between the three factors, especially the impact of safety climate on safety behavior, which can help to develop more comprehensive and systematic safety management measures and improve the safety performance of construction workers.

6. Conclusions

This study explored the impacts of the demographic factors of gender, age, work experience, education level, and employer size on safety climate and construction worker safety behavior. The site survey was conducted at Chinese construction projects, and 536 valid questionnaires were collected from frontline construction workers. Five-point Likert scales were adopted to assess safety climate and construction worker safety behavior. The method of *t*-test was used to test the worker safety differences by gender, while one-way ANOVA was adopted to examine the worker safety differences related to the effects of four other demographic factors (age, work experience, education level, and employer size). The sub-group analysis helps identify whether a specific demographic factor significantly influences the sub-dimensions of safety climate and safety behavior and shows what the sub-group differences are under the effects of the demographic factor.

The results reveal that: (1) all five demographic factors (age, gender, education level, work experience, and employer size) have significant impacts on certain components of safety climate, while three (age, education level, and employer size) have significant influences on certain components of construction worker safety behavior; (2) gender and education level significantly correlate with coworker perception; (3) age and work experience significantly impact management commitment; (4) employer size significantly affects five sub-dimensions of safety climate; (5) two demographic factors (gender and work experience) do not show significant effects on construction worker safety behavior. The results remind the safety management personnel to take a pragmatic approach and to formulate safety measures based on the demographic characteristics of different worker sub-groups.

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Institutional Review Board Statement: Ethical review and approval were dropped for this study since it did not involve direct intervention of the subjects.

Informed Consent Statement: Ethical review and approval were waived for this study since the survey was anonymous, and the respondents agreed that researchers use their answers/opinions for analysis.

Data Availability Statement: The data that support the findings of this study are available from the corresponding author.

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Appendix A. The Questionnaire

1. Demographic information	
Employer size	☐1–20 employees ☐21–300 employees ☐300 or more employees
Gender	☐Male ☐Female
Age	☐<30 years ☐30–39 years ☐40–49 years ☐≥50 years
Education level	☐Elementary school ☐Middle school ☐High school or higher
Work experience	☐<5 years ☐5–9 years ☐10–14 years ☐≥15 years

2. Safety climate

Items	**Strongly Disagree**	**Disagree**	**Uncertain**	**Agree**	**Strongly Agree**
Management commitment					
Our management provides enough safety training programs	1	2	3	4	5
Our management conducts frequent safety inspections	1	2	3	4	5
Our management provides safe equipment	1	2	3	4	5
Our management is strict about working safely when work falls behind schedule	1	2	3	4	5
Our management gives safety personnel the power they need to do their job	1	2	3	4	5
After an unsafety event, our management focuses on how to solve problems and improve safety, rather than seeking to pin blame on specific individuals	1	2	3	4	5
Supervisor perception					
My supervisor spends time showing me the safest way to do things at work	1	2	3	4	5
My supervisor expresses satisfaction when I perform my job safely	1	2	3	4	5
My supervisor talks about values and beliefs in the importance of safety	1	2	3	4	5
My supervisor makes sure that we receive appropriate rewards for achieving safety targets on the job	1	2	3	4	5
My supervisor behaves in a way that displays a commitment to a safe workplace	1	2	3	4	5
My supervisor keeps workers informed of safety rules	1	2	3	4	5
Coworker perception					
My coworkers ignore safety rules (R)	1	2	3	4	5
My coworkers encourage others to be safe	1	2	3	4	5
My coworkers take chances with safety (R)	1	2	3	4	5
My coworkers keep work area clean	1	2	3	4	5

Work pressure					
There are enough workers to carry out the required work (R)	1	2	3	4	5
There is sufficient "thinking time" to enable workers to plan and carry out the required work (R)	1	2	3	4	5
Role overload					
I am so busy on the job that I can't take normal breaks	1	2	3	4	5
There is too much work to do in my job for it all to be done well	1	2	3	4	5
3. Safety behavior					
Items	**Strongly Disagree**	**Disagree**	**Uncertain**	**Agree**	**Strongly Agree**
Safety compliance					
I use all the necessary safety equipment to do my job	1	2	3	4	5
I use the correct safety procedures for carrying out my job	1	2	3	4	5
I ensure the highest levels of safety when I carry out my job	1	2	3	4	5
Safety participation					
I promote the safety program within the organization	1	2	3	4	5
I put in extra effort to improve the safety of the workplace	1	2	3	4	5
I voluntarily carry out tasks or activities that help to improve workplace safety	1	2	3	4	5

Note: R stands for the reverse items.

Appendix B. The Subgroup Analysis Tables

Table A1. Impact of gender on safety climate.

Latent Variable	Judgment of Variance Homogeneity			Judgment of Mean Difference		
	F	**Sig.**	**Variance Is Homogeneous**	**T**	**Sig. (Two-Tailed)**	**Difference Is Significant**
Management commitment	0.008	0.931	Yes	0.428	0.669	No
Supervisor perception	2.664	0.103	Yes	0.754	0.451	No
Coworker perception	4.053	0.045	No	2.032	0.047	Yes
Work pressure	0.739	0.390	Yes	−1.837	0.067	No
Role overload	6.640	0.010	No	−1.768	0.082	No

Note: F = Levene F statistic; T = t statistic.

Table A2. Sub-group comparison of coworker perception for gender.

Latent Variable	Gender	N	Mean	S.D.	Mean of S.E.
Coworker perception	Male	491	3.766	0.713	0.032
	Female	45	3.583	0.564	0.084

Note: N = number of workers; S.D. = standard deviation; S.E. = standard error.

Table A3. Impact of age on safety climate.

Latent Variable	Judgment of Variance Homogeneity			Judgment of Mean Difference		
	F	Sig.	Variance Is Homogeneous	F	Sig. (Two-Tailed)	Difference Is Significant
Management commitment	2.159	0.092	Yes	4.544	0.004	Yes
Supervisor perception	0.586	0.624	Yes	1.787	0.149	No
Coworker perception	0.893	0.445	Yes	1.006	0.390	No
Work pressure	1.467	0.223	Yes	0.384	0.765	No
Role overload	0.739	0.529	Yes	1.086	0.355	No

Note: F = Levene F statistic.

Table A4. Sub-group comparison of management commitment for age.

Latent Variable	Method	(I) Age	(J) Age	(I-J) Average Difference	S.E.	Sig.
Management commitment	Scheffe	≥50	<30	−0.340 *	0.106	0.017
			30–39	−0.236	0.088	0.069
			40–49	−0.120	0.091	0.629

Note: S.E. = standard error; * stands for significant differences.

Table A5. Impact of education level on safety climate.

Latent Variable	Judgment of Variance Homogeneity			Judgment of Mean Difference		
	F	Sig.	Variance Is Homogeneous	F	Sig. (Two-Tailed)	Difference Is Significant
Management commitment	0.071	0.932	Yes	2.067	0.128	No
Supervisor perception	0.011	0.989	Yes	1.232	0.293	No
Coworker perception	2.019	0.134	Yes	3.755	0.024	Yes
Work pressure	2.516	0.082	Yes	1.477	0.229	No
Role overload	1.472	0.230	Yes	0.226	0.797	No

Note: F = Levene F statistic.

Table A6. Sub-group comparison of coworker perception for education level.

Latent Variable	Method	(I) Education Level	(J) Education Level	(I-J) Average Difference	S.E.	Sig.
Coworker perception	Scheffe	High school or higher	Elementary school	0.227 *	0.088	0.038
			Middle school	0.163	0.073	0.083

Note: S.E. = standard error; * stands for significant differences.

Table A7. Impact of work experience on safety climate.

Latent Variable	Judgment of Variance Homogeneity			Judgment of Mean Difference		
	F	Sig.	Variance Is Homogeneous	F	Sig. (Two-Tailed)	Difference is Significant
Management commitment	0.566	0.638	Yes	3.881	0.009	Yes
Supervisor perception	0.577	0.631	Yes	2.220	0.085	No
Coworker perception	0.598	0.617	Yes	2.350	0.072	No
Work pressure	1.170	0.321	Yes	1.443	0.229	No
Role overload	2.682	0.046	No	1.580	0.193	No

Note: F = Levene F statistic.

Table A8. Sub-group comparison of management commitment for work experience.

Latent Variable	Method	(I) Work Experience	(J) Work Experience	(I-J) Average Difference	S.E.	Sig.
Management commitment	Scheffe	≥15 years	<5 years	0.036	0.090	0.983
			5–9 years	0.232	0.084	0.055
			10–14 years	0.147	0.090	0.448

Note: S.E. = standard error.

Table A9. Impact of employer size on safety climate.

Latent Variable	Judgment of Variance Homogeneity			Judgment of Mean Difference		
	F	Sig.	Variance Is Homogeneous	F	Sig. (Two-Tailed)	Difference Is Significant
Management commitment	9.452	0.000	No	7.943	0.000	Yes
Supervisor perception	11.534	0.000	No	5.081	0.007	Yes
Coworker perception	2.742	0.065	Yes	4.569	0.011	Yes
Work pressure	7.872	0.000	No	6.705	0.001	Yes
Role overload	8.124	0.000	No	5.301	0.005	Yes

Note: F = Levene F statistic.

Table A10. Sub-group comparison of safety climate for employer size.

Latent Variable	Method	(I) Employer Size	(J) Employer Size	(I-J) Average Difference	S.E.	Sig.
Management commitment	Tamhance's T2	>300 people	1–20 people	0.360 *	0.119	0.009
			21–300 people	0.105	0.056	0.178
Supervisor perception	Tamhance's T2	>300 people	1–20 people	0.307 *	0.125	0.047
			21–300 people	0.147 *	0.057	0.033
Coworker perception	Scheffe	>300 people	1–20 people	0.308 *	0.102	0.011
			21–300 people	0.140	0.074	0.170
Work pressure	Tamhance's T2	>300 people	1–20 people	−0.219	0.111	0.143
			21–300 people	−0.261 *	0.063	0.000
Role overload	Tamhance's T2	>300 people	1–20 people	−0.213	0.175	0.535
			21–300 people	−0.359 *	0.113	0.005

Note: S.E. = standard error; * stands for significant differences.

Table A11. Impact of age on safety behavior.

Latent Variable	Judgment of Variance Homogeneity			Judgment of Mean Difference		
	F	Sig.	Variance Is Homogeneous	F	Sig. (Two-Tailed)	Difference Is Significant
Safety compliance	1.449	0.228	Yes	6.583	0.000	Yes
Safety participation	6.925	0.000	No	4.543	0.004	Yes

Note: F = Levene F statistic.

Table A12. Sub-group comparison of safety behavior for age.

Latent Variable	Method	(I) Age	(J) Age	(I-J) Average Difference	S.E.	Sig.
Safety compliance	Scheffe	40–49	<30	−0.398 *	0.103	0.002
			30–39	−0.261 *	0.075	0.008
			≥50	−0.255	0.108	0.136
Safety participation	Tamhance's T2	40–49	<30	−0.330 *	0.123	0.047
			30–39	−0.048	0.101	0.998
			≥50	−0.416 *	0.114	0.002

Note: S.E. = standard error; * stands for significant differences.

Table A13. Impact of education level on safety behavior.

Latent Variable	Judgment of Variance Homogeneity			Judgment of Mean Difference		
	F	Sig.	Variance Is Homogeneous	F	Sig. (Two-Tailed)	Difference Is Significant
Safety compliance	3.861	0.022	No	2.101	0.123	No
Safety participation	7.189	0.001	No	7.638	0.001	Yes

Note: F = Levene F statistic.

Table A14. Sub-group comparison of safety participation for education level.

Latent Variable	Method	(I) Education Level	(J) Education Level	(I-J) Average Difference	S.E.	Sig.
Safety participation	Tamhance's T2	High school or higher	Elementary school	0.342 *	0.109	0.006
			Middle school	0.384 *	0.096	0.000

Note: S.E. = standard error; * stands for significant differences.

Table A15. Impact of employer size on safety behavior.

Latent Variable	Judgment of Variance Homogeneity			Judgment of Mean Difference		
	F	Sig.	Variance Is Homogeneous	F	Sig. (Two-Tailed)	Difference Is Significant
Safety compliance	5.993	0.003	No	3.110	0.045	Yes
Safety participation	15.041	0.000	No	4.011	0.019	Yes

Note: F = Levene F statistic.

Table A16. Sub-group comparison of safety behavior for employer size.

Latent Variable	Method	(I) Employer Size	(J) Employer Size	(I-J) Average Difference	S.E.	Sig.
Safety compliance	Tamhance's T2	1–20 people	21–300 people	−0.175	0.119	0.377
			>300 people	−0.276	0.128	0.097
Safety participation	Tamhance's T2	1–20 people	21–300 people	−0.348	0.145	0.055
			>300 people	−0.272	0.169	0.296

Note: S.E. = standard error.

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